

Interface

File: Interface.kicad_sch

Power

File: Power.kicad_sch

ESP32-S3

File: ESP32-S3.kicad_sch

IO Usage Table

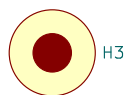
ESP32-S3 USB : IO19 IO20
SPI FLASH : IO30~IO35
Battery Level Check : IO18
A : IO46
B : IO45
WS2812 : IO41
Audio Codec : IO7 IO8 IO9 IO10 IO11 IO12 IO13
LCD240 : IO14 IO15 IO16 IO17 IO21
PH2.0-Port : IO5 IO6
SD CARD : IO38 IO39 IO40
Extended IO : IO1 IO2 IO3 IO4 IO42 IO43 IO44 IO48



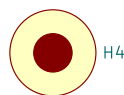
H1



H2



H3



H4

[www.github.com/nulllaborg/esp32-s3-audio-dev-board](https://github.com/nulllaborg/esp32-s3-audio-dev-board)
https://docs.emakefun.com/#/zh-cn/esp32/ai-vox3/ai-vox3_zh-cn

Sheet: /
File: AI-VOX3.kicad_sch

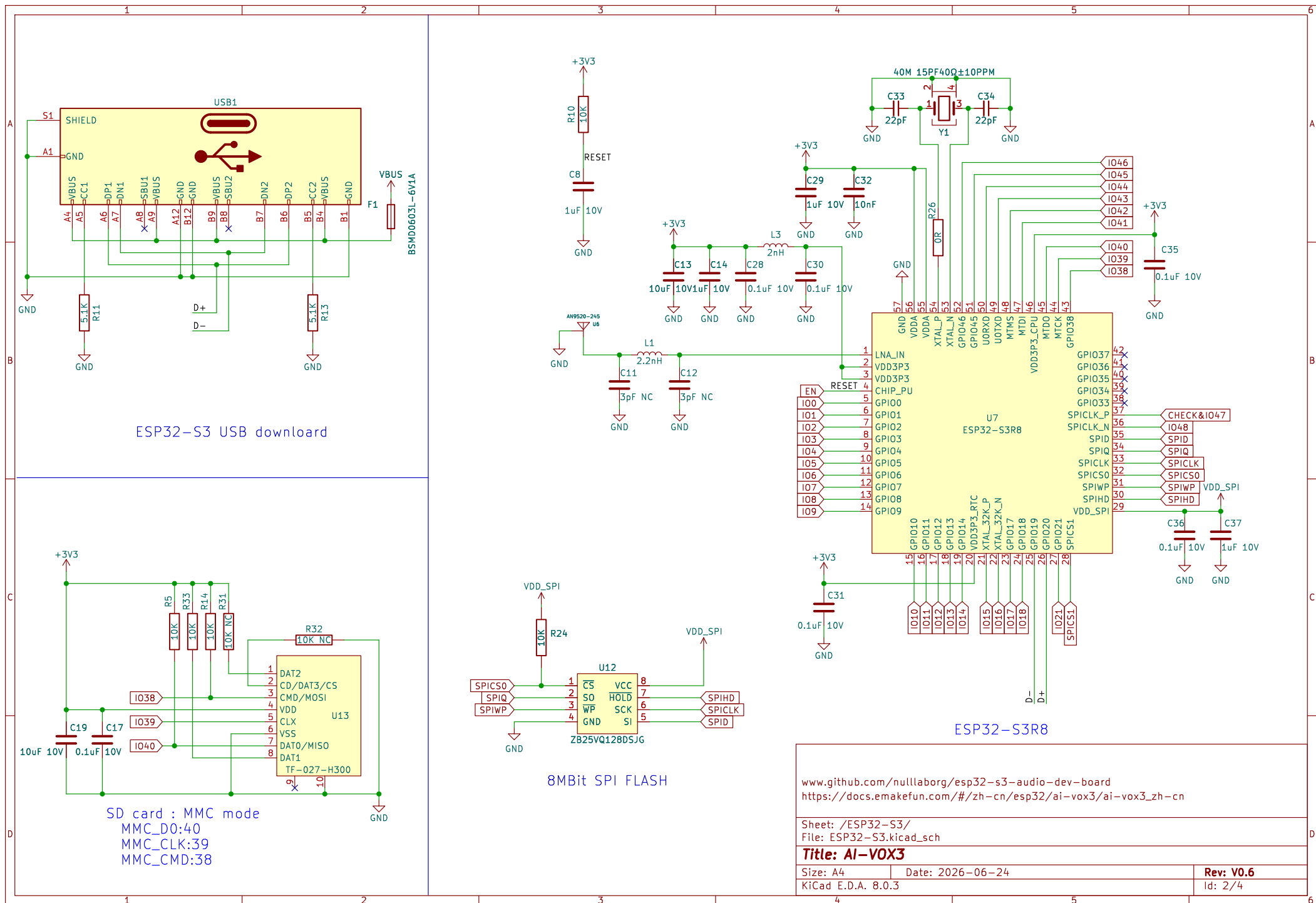
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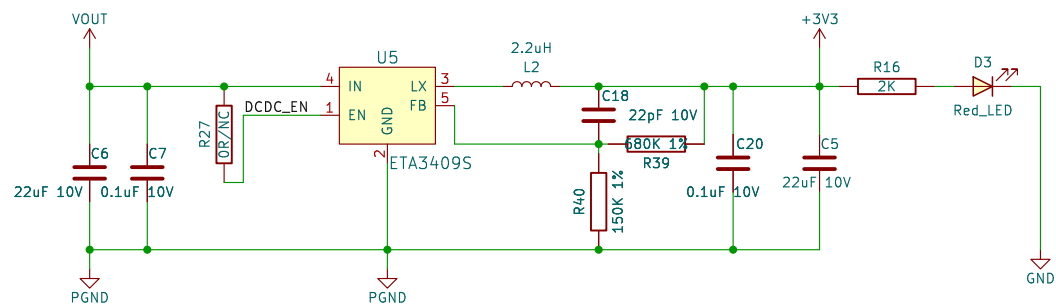
Size: A4 Date: 2026-04-27

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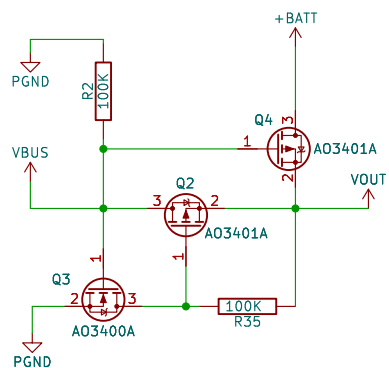
Rev: V1.1

Id: 1/4

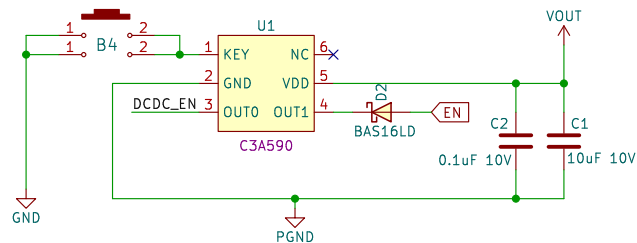




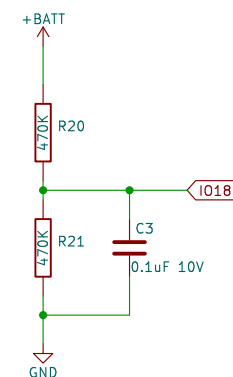
DC:DC VOUT->3.3V 2A



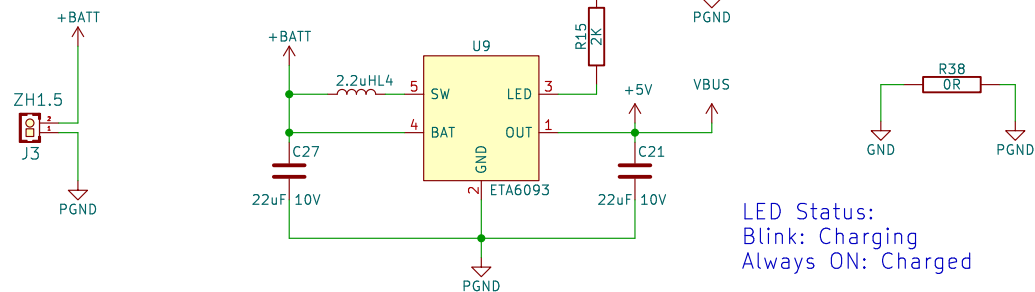
USB and Li-battery power switch



Power Management
Short press <= 300ms: Turn ON, red power led on
Long press > 500ms: Turn OFF, red power led off
Power ON: short press <= 300ms reset esp32-s3



Battery Level Check
 $ADC_value = 1/2 * V_{bat}$



LED Status:
Blink: Charging
Always ON: Charged

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https://docs.emakefun.com/#/zh-cn/esp32/ai-vox3/ai-vox3_zh-cn

Sheet: /Power/
File: Power.kicad_sch

Title: AI-VOX3

Size: A4 Date: 2025-05-14

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Rev: V0.6

Id: 3/4

